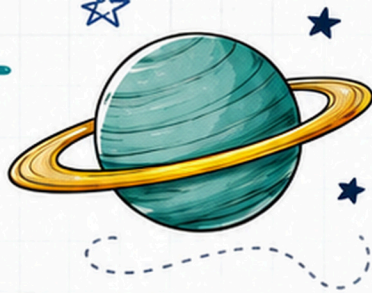


SAVE TIME. STAY PREPARED.
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EMERGENCY SUB PLANS

$E=mc^2$

MIDDLE SCHOOL SCIENCE

VOLUME 2

10
NEW
LESSONS

Ten more no-prep emergency lessons for the day you cannot be there. **No lab. No equipment.**



NO-PREP –
Just Print
& Go!



READY-TO-USE
Complete Plans
& Materials



FOR BUSY DAYS –
Engaging,
Meaningful,
Easy to Follow

QUALITY
RESOURCES.
CONFIDENT
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- ✓ 10 brand new lessons, none repeated from Volume 1
- ✓ No lab, no equipment, no science background needed
- ✓ Every lesson has the words to say and a timing table
- ✓ Runs entirely on paper and a pencil
- ✓ Full answer keys that explain why, not just what



* Created by
teachers,
for teachers!



START HERE

Ten complete emergency science lessons. Each runs for a full class period, needs nothing but photocopies, and is written so a substitute with no science background can teach it confidently.

This is Volume 2. All ten lessons are new. Nothing here repeats Volume 1, so the two packs sit side by side in the drawer and give you twenty emergency days.

What every lesson contains

PAGE	WHAT IT IS	WHO IT IS FOR
1	Teacher briefing. Timing table, the exact words to say, the three questions students will ask with the answers written out, and what to do with early finishers.	The substitute. Keep this page.
2	Reading or task page. The material for the lesson, with numbered paragraphs and any diagram needed.	Copy for students.
3	Question page. Six to nine questions with writing space built in.	Copy for students.
4	Go further page. Extension task and a challenge for students who finish early.	Copy for students.
5	Answer key. Every answer explained, plus marking notes and the wrong answers to expect.	The substitute. Keep this page.

ABSOLUTELY NO LAB, AND THAT IS THE POINT

Most schools will not let a substitute run a practical, and rightly so. Equipment and safety supervision need the regular teacher present. That is exactly why science sub plans are hard to find.

Nothing in this pack is heated, poured, cut or plugged in. Every lesson runs on paper. The quick start page says so plainly, and each opening script pre-empts the inevitable question about doing an experiment with a reason that is actually true.

Reading level and grouping

Passages sit in the grades 6 to 8 band. Every lesson works as independent silent work, as partner work, or read aloud by the substitute. The briefing page tells the substitute how to run all three, so no decision is left to guesswork on the day.

Substitute Quick Start

FOR THE SUBSTITUTE

If you have five minutes before the bell, read this page and nothing else. It is enough.

IN THE FIRST FIVE MINUTES

1. Find the lesson the teacher left, or pick any lesson from the index. They are independent. Order does not matter.
2. Copy the three student pages, one set per student.
3. Keep the briefing page and the answer key. Do not photocopy those.
4. Read the briefing page. It has a timing table and the words to say.

DO NOT OPEN THE CUPBOARDS

Nothing in this pack needs equipment, and you should not take anything out of a science storeroom, fridge or cupboard. If a student tells you the class always does a practical on this day, the honest answer is that safety supervision needs their regular teacher present.

How the period runs, every time

MINUTES	WHAT HAPPENS
0-5	Students settle. You read the opening lines from the briefing page out loud. Hand out the student pages.
5-15	Students read the task page. Silent, partner, or read aloud by you. Your choice.
15-40	Students work through the question page. You circulate.
40-50	Anyone finished moves to the Go Further page. Collect everything with two minutes to go.

IF A STUDENT ASKS YOU SOMETHING YOU CANNOT ANSWER

Check the briefing page first. The three most common questions for that lesson are already answered there.

If it is not there, this always works and is always true: *"Write down your best answer and put a star next to it. Your teacher will look at the starred ones."*

Lesson Index

ALL TEN LESSONS IN VOLUME 2

Lessons are numbered 11 to 20 so they sit alongside Volume 1 without clashing. Every lesson is independent. Teach them in any order.

#	LESSON	FOCUS	NGSS	TIME
11	Something Changed	Physical vs chemical change, conservation of mass	MS-PS1-2	40-50 min
12	Good Vibrations	Waves: amplitude, wavelength, frequency	MS-PS4-1	40-50 min
13	Where Traits Come From	Heredity, alleles, Punnett squares	MS-LS3-2	45-55 min
14	The Beetles That Survived	Natural selection and adaptation	MS-LS4-4	45-55 min
15	Trading Gases	Photosynthesis and cellular respiration	MS-LS1-6	40-50 min
16	The Delivery Network	Circulatory and respiratory systems	MS-LS1-3	40-50 min
17	Why It Floats	Density and buoyancy	MS-PS1-1	45-55 min
18	Closing the Loop	Circuits, series and parallel	MS-PS3-5	40-50 min
19	Same Moon, Different Shape	Moon phases and the seasons	MS-ESS1-1	45-55 min
20	Reading the Rocks	Rock cycle, strata, relative dating	MS-ESS1-4	45-55 min

WHICH LESSON SHOULD I LEAVE?

Planned absence, want real work done: Lesson 13 or 17. Both produce calculation you can grade.

Emergency, class you do not know well: Lesson 11 or 16. Both are self-contained with a low frustration ceiling.

Restless group, last period: Lesson 14 or 19. Both have a strong hook and a visible finish line.

Out several days: 15 then 16 run well as a pair, and 19 then 20 do the same for earth and space.

NGSS codes are given as the closest matching middle school performance expectation. This resource is independently created and is not affiliated with or endorsed by any state agency or standards body.

Something Changed

TEACHER BRIEFING • KEEP THIS PAGE

NO SCIENCE BACKGROUND NEEDED. NO LAB, NO EQUIPMENT.

Everything you need is on this page and the answer key. Nothing in this lesson is heated, poured, cut or plugged in.

Copy for students: the next three pages. **Keep:** this page and the answer key.

What this lesson does

Students sort eleven everyday changes into physical and chemical, using evidence rather than memory. The lesson deliberately includes three cases that look chemical and are not, because that is where the real understanding sits. It closes on conservation of mass, which is the idea that makes the whole topic make sense.

How the period runs

MIN	WHAT YOU DO
0-5	Read the opening lines below. Hand out all three student pages.
5-14	Students read the page. Partner reading works well.
14-38	Question page. The sorting table is the main task.
38-48	Go Further: the sealed jar problem.
48-50	Collect all pages.

SAY THIS TO START

Here is the question for today. When you toast bread, where does the bread go? It gets darker, it gets lighter in weight, it smells different, and you can never turn it back. Compare that to melting an ice cube, which you can undo any time you like. Both are changes. Only one of them made something new.

The three questions students will ask

THEY ASK	YOU SAY
"Is dissolving sugar a chemical change?"	No. The sugar is still sugar, it is just spread through the water. You can get it back by evaporating the water. That is the test.
"Why does burnt toast weigh less?"	Because gases and water vapor left the bread and went into the air. Nothing was destroyed. It just left.
"Is cooking an egg reversible?"	No, and that is the giveaway. You cannot un-cook an egg. New substances formed when the proteins changed.

FINISHED EARLY?

The Go Further page asks what a sealed jar would weigh before and after steel wool inside it rusts. It is the conservation of mass idea in one concrete question.

Read: New Stuff, or the Same Stuff?

BEFORE YOU READ

There is one question that sorts almost every change you will ever meet: **did a new substance get made?** Everything below is about how to tell.

New Stuff, or the Same Stuff?

Written for this lesson

- 1 Break a plate and you have smaller pieces of plate. Melt an ice cube and you have the same water in a different shape. Tear paper, dissolve salt, boil a kettle, hammer a nail flat. In every one of these, the stuff is still the stuff. It changed shape, size or state, and nothing new was made. These are **physical changes**.
- 2 Now burn the paper. What you are left with is not smaller paper. It is ash, smoke and a lot of warm air, and no amount of effort will turn it back. Something genuinely new exists that did not exist before. That is a **chemical change**.
- 3 Scientists look for specific evidence. A chemical change usually announces itself in at least one of these ways: a gas appears where there was none, a solid forms out of two clear liquids, the color changes permanently, light is given off, or heat is released or absorbed without anything heating it.
- 4 None of those is proof on its own, which is where people get caught. Boiling water produces a gas. Nothing new was made, because that gas is still water. A tomato ripening from green to red changes color, and that one is chemical. The evidence is a clue, not a verdict. The verdict comes from the real test.
- 5 **The real test is whether you can get back what you started with, easily.** Freeze the melted ice and you have the ice cube back. Evaporate the salt water and the salt is sitting there waiting. But you cannot un-burn paper, un-rust a nail or un-cook an egg. Those took a one-way trip.
- 6 This raises a question that bothered chemists for two hundred years. If you burn a log and the ash weighs far less than the log did, where did the missing mass go? For a long time the leading answer was that burning released a substance called phlogiston out of the wood, and that things which burn are simply full of it.
- 7 The problem showed up when people burned metals. Burn magnesium and the powder left behind weighs *more* than the metal did. If burning released something, the mass should have gone down. Instead it went up. Phlogiston would have to weigh less than nothing.
- 8 The answer turned out to be much simpler. Nothing is ever lost. When a log burns, gases and water vapor leave and go into the air, which is why the ash is lighter. When magnesium burns, oxygen from the air joins the metal, which is why the powder is heavier. Weigh everything, including the air, and the total never changes.
- 9 This is the **conservation of mass**: in any change, physical or chemical, the total mass of everything you started with equals the total mass of everything you end with. Atoms are not created and not destroyed. They are only rearranged. The reason a change ever appears to gain or lose mass is that something invisible walked in or out of the room.

Questions: Something Changed

Sort each change. Write **P** for physical or **C** for chemical, then give the reason. Three of these are trickier than they look.

#	THE CHANGE	P OR C	HOW YOU CAN TELL
1	Ice melting in a glass		
2	A nail rusting over a winter		
3	Sugar dissolving in tea		
4	Toasting a slice of bread		
5	Boiling water into steam		
6	A green tomato turning red		
7	Chopping wood into kindling		
8	Fireworks exploding		
9	Milk going sour in a warm car		
10	Hammering copper into a sheet		
11	A glowstick being snapped		

12 Paragraph 7 says burning magnesium leaves powder that weighs more than the metal did. Explain where the extra mass came from.

13 A log weighs 4.0 kg. After burning, the ash weighs 0.1 kg. Has mass been destroyed? Explain in one or two sentences.

Go Further: The Sealed Jar

THE PROBLEM

A scientist puts a piece of damp steel wool inside a glass jar, seals the lid tightly so nothing can get in or out, and weighs the whole sealed jar. It comes to **212.0 grams**.

She leaves it on a shelf for three weeks. The steel wool rusts badly. She weighs the sealed jar again.

1 What does the sealed jar weigh after three weeks? Circle one and explain your reasoning.

More than 212.0 g / Exactly 212.0 g / Less than 212.0 g

2 Now she opens the jar, takes out just the rusted steel wool and weighs it on its own. Compared with the steel wool she started with, is it heavier, lighter or the same? Why?

THE TRICKY BIT

Those two answers are different, and both are correct. Explain in your own words why the sealed jar and the steel wool on its own do not give the same result.

CHALLENGE

For two hundred years the accepted explanation for burning was phlogiston, a substance supposedly released by anything that burns. Burning metals gaining mass should have killed the idea immediately. Why do you think scientists held onto it for so long instead? Two or three sentences.

Answer Key: Something Changed

SORTING TABLE

#		WHY
1	P	Still water, just a different state. Refreeze it.
2	C	Rust is a new substance. Cannot be undone easily.
3	P	Tricky one. Still sugar, just spread out. Evaporate the water and it returns.
4	C	New substances formed. Cannot un-toast bread.
5	P	Tricky one. A gas appears, but that gas is still water. Condense it back.
6	C	Tricky one. Color change here is a real chemical reaction inside the fruit. It never goes back.
7	P	Smaller pieces of the same wood.
8	C	Light, heat, gas, and no way back.
9	C	Bacteria produce new substances. Irreversible.
10	P	Still copper, different shape.
11	C	Snapping mixes two chemicals that react and give off light. The glow fades and cannot be reset.

Items 3, 5 and 6 are the deliberate traps. Students almost always mark dissolving and boiling as chemical because something visibly happens, and mark the tomato as physical because it is just a color. Worth calling out to the class if there is time.

Mark the reason column, not the letter. A student who writes P for item 6 but reasons carefully has learned more than one who guessed C.

12. Oxygen from the air joined the magnesium and became part of the powder. Mass was added from the surrounding air, not created.

13. No. Gases and water vapor left the log and went into the air. If you could weigh the ash plus every gas released, the total would still be 4.0 kg.

GO FURTHER PAGE

1. Exactly 212.0 g. The jar is sealed, so nothing can enter or leave. Rusting rearranges atoms already inside the jar. Total mass is unchanged.

The most common wrong answer is "less, because rust is flaky." Ask them what could possibly have left a sealed jar.

2. Heavier. The steel wool gained oxygen atoms that came from the air inside the jar.

The tricky bit: The jar total stays the same because the oxygen simply moved from the air inside the jar into the steel wool. The steel wool alone gets heavier because it gained that oxygen. Mass moved within the system; none was created or destroyed.

This is the conceptual heart of the lesson. Full credit needs the idea that mass moved from one part of the sealed system to another.

Challenge: Open. Strong answers: phlogiston explained most everyday burning perfectly well; scientists adjusted the theory rather than abandoning it; there was no better alternative until gases could be weighed accurately.

Any answer engaging with why a useful-but-wrong idea survives gets credit.

Good Vibrations

TEACHER BRIEFING · KEEP THIS PAGE

NO SCIENCE BACKGROUND NEEDED. NO LAB, NO EQUIPMENT.

Everything you need is on this page and the answer key. Nothing in this lesson is heated, poured, cut or plugged in.

Copy for students: the next three pages. **Keep:** this page and the answer key.

What this lesson does

Students read three labeled waves off a single diagram and connect the measurements to things they can actually hear and see: loudness, pitch and color. The lesson targets the most common misconception in the topic, which is that a taller wave is a higher note.

How the period runs

MIN	WHAT YOU DO
0-5	Read the opening lines below. Hand out all three student pages.
5-13	Students read the page and study the wave diagram.
13-37	Question page. Questions 1 to 4 read directly off the diagram, so they should move quickly.
37-48	Go Further: the sound and light race.
48-50	Collect all pages.

SAY THIS TO START

Everything you hear and everything you see arrives as a wave. The surprising part is that only two measurements decide almost all of it. How tall the wave is decides how loud a sound is, or how bright a light is. How squeezed together the waves are decides the pitch of the sound, or the color of the light. Two numbers. That is nearly the whole topic.

The three questions students will ask

THEY ASK	YOU SAY
"Does a bigger wave mean a higher note?"	No, and this is the classic mix-up. A bigger wave is a louder note at the same pitch. Pitch depends on how many waves pass per second.
"Why do you see lightning before you hear thunder?"	Light travels roughly a million times faster than sound, so the light arrives almost instantly while the sound takes about 5 seconds per mile.
"Can sound travel through space?"	No. Sound needs particles to vibrate. Space is close to empty, so there is nothing to carry it. Light does not need particles.

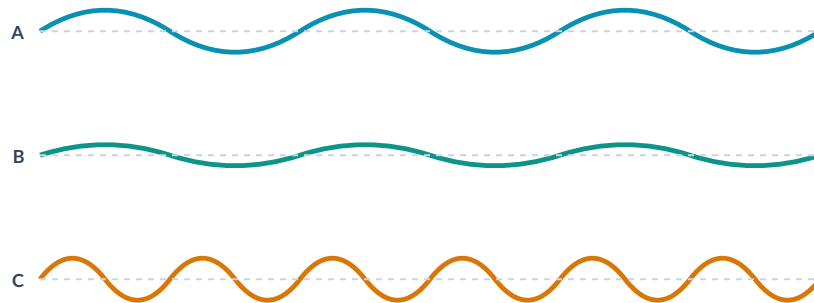
FINISHED EARLY?

The Go Further page has a timing problem about a storm and a fireworks display, using the 5 seconds per mile rule.

Read: Reading a Wave

BEFORE YOU READ

Study the three waves below carefully. Every question on the next page depends on being able to read them. All three are drawn to the same scale.



All three drawn to the same scale

- 1 A wave is a way of moving energy from one place to another without moving the stuff in between very far. Drop a stone in a pond and the ripple travels outward, but a leaf floating on the surface just bobs up and down. The energy travels. The water does not.
- 2 Two measurements describe almost any wave. The **amplitude** is how far it swings from the middle line, in other words how tall it is. The **wavelength** is the distance from one peak to the next. Closely related to wavelength is the **frequency**, which counts how many complete waves pass a fixed point each second. Short wavelength means the waves are packed close together, so more of them go past every second, so the frequency is high.
- 3 For sound, these two measurements do two different jobs. Amplitude controls **loudness**: a bigger swing carries more energy and hits your eardrum harder. Frequency controls **pitch**: more waves per second is a higher note. Turning up the volume on a song makes the amplitude larger. It does not make the singer sound like a chipmunk, because the frequency has not changed.
- 4 For light, the same two measurements do the equivalent jobs. Amplitude controls **brightness**. Wavelength controls **color**. Red light has the longest wavelength we can see and violet the shortest, with the rest of the rainbow in between. Beyond red is infrared, which we feel as warmth. Beyond violet is ultraviolet, which we do not see at all but which burns skin.
- 5 There is one important difference between the two. Sound is a vibration passed from particle to particle, so it needs something to travel through: air, water, steel, anything with particles. Take the particles away and sound stops dead. Light needs nothing at all, which is why sunlight crosses ninety-three million miles of near-empty space to reach us while the same distance would be completely silent.
- 6 They also travel at very different speeds. Sound moves through air at about **5 seconds per mile**. Light moves at 186,000 miles per second, so over any distance on Earth it is effectively instant. That is why you see lightning and then wait for the thunder.

Questions: Good Vibrations

Questions 1 to 4 refer to waves **A**, **B** and **C** on the reading page.

1 Which two waves have the same wavelength as each other? Which one is different?

2 Which two waves have the same amplitude? Which one is different?

3 If all three are sound waves, which would be the quietest, and which would be the highest note? Explain both.

4 Wave C has twice as many waves in the same distance as wave A. What does that tell you about C's frequency compared with A's?

5 You turn the volume up on a speaker. Which measurement changes, and which stays the same?

6 Explain why an astronaut on the Moon cannot hear an explosion happening a hundred feet away, but can see it clearly.

7 Infrared and ultraviolet are both invisible to us. Which one has a longer wavelength than red light, and which has a shorter wavelength than violet?

Go Further: Counting the Gap

THE RULE

Light arrives effectively instantly. Sound takes about **5 seconds to travel one mile**. So the gap between seeing something and hearing it tells you how far away it is.

SITUATION	GAP YOU COUNT	HOW FAR AWAY?
Lightning flash, then thunder	15 seconds	
Second lightning flash, then thunder	5 seconds	
Firework burst, then the bang	2 seconds	
Starting pistol at a track meet	0.5 seconds	

1 Between the first flash and the second, is the storm moving toward you or away? How do you know?

2 At a track meet the timekeeper stands at the finish line and starts the clock when they hear the starting pistol, 0.5 seconds away. Does this make the recorded times too fast or too slow? By how much?

CHALLENGE

Sound travels faster through water than through air, and faster still through steel. Using the idea that sound is a vibration passed from particle to particle, suggest why. Think about how close together the particles are.

IF YOU STILL HAVE TIME

A bat hunts by sending out very high frequency sound and listening for the echo. Bats need to detect small insects. Explain why a short wavelength is better than a long one for finding something small.

Answer Key: Good Vibrations

QUESTION PAGE

1. A and B have the same wavelength (three waves across the same distance). C is different, with six waves in that distance, so its wavelength is half theirs.
2. A and C have the same amplitude. B is half as tall.
3. **Quietest: B**, because it has the smallest amplitude. **Highest note: C**, because it has the shortest wavelength and therefore the highest frequency.
This question is the whole lesson. The classic error is naming C as loudest or A as highest. If a student answers "B is quietest and B is highest", they have merged the two ideas. Point at the diagram and ask which one is taller and which one is more squeezed up.
4. C's frequency is twice A's. More waves in the same distance means more pass a fixed point each second.
5. Amplitude increases. Frequency and wavelength stay the same, which is why the pitch does not change.
6. Sound needs particles to pass the vibration along. The Moon has essentially no atmosphere, so there is nothing to carry the sound. Light needs no particles, so it crosses the gap normally.
7. Infrared has a longer wavelength than red. Ultraviolet has a shorter wavelength than violet.

GO FURTHER PAGE

GAP	DISTANCE	WORKING
15 s	3 miles	15 / 5
5 s	1 mile	5 / 5
2 s	0.4 miles	2 / 5
0.5 s	0.1 miles, about 530 feet	0.5 / 5

1. Toward you. The gap shrank from 15 seconds to 5 seconds, so the storm went from 3 miles away to 1 mile away.
2. **Too fast.** The clock starts 0.5 seconds late, so every recorded time is 0.5 seconds shorter than the real one. This is why real track timing is triggered electrically at the pistol, not by someone listening.

Strong students spot that 0.5 seconds is enormous in a sprint. Worth noting.

Challenge: In water and especially in steel the particles are much closer together, so a vibration is passed along faster. In a gas the particles are far apart and have to travel further before colliding.

Bats: A wavelength longer than the insect tends to pass around it without a useful echo. A short wavelength reflects off small objects, so it can locate something small.

Do not require the word diffraction. "Long waves go around small things" is a full-credit answer at this level.

Where Traits Come From

TEACHER BRIEFING · KEEP THIS PAGE

NO SCIENCE BACKGROUND NEEDED. NO LAB, NO EQUIPMENT.

Everything you need is on this page and the answer key. Nothing in this lesson is heated, poured, cut or plugged in.

Copy for students: the next three pages. **Keep:** this page and the answer key.

What this lesson does

Students learn to fill a Punnett square and read ratios off it. The worked example is done for them on the reading page, so no student is stuck at the start. The lesson closes on the point that a ratio is a probability for each child separately, not a guarantee about a family, which is the misconception that survives most teaching of this topic.

How the period runs

MIN	WHAT YOU DO
0-6	Read the opening lines below. Hand out all three student pages. Walk through the worked square on page 1 out loud.
6-14	Students read the page.
14-40	Question page. Students draw and fill three squares.
40-53	Go Further: the four-child problem.
53-55	Collect all pages.

SAY THIS TO START

You have two copies of nearly every gene, one from each parent. Sometimes those two copies disagree. When they do, one usually wins, and the one that wins is called dominant. Today you are going to work out, on paper, what the odds are for a child, using a grid that takes about twenty seconds to draw.

The three questions students will ask

THEY ASK	YOU SAY
"What is the difference between a gene and an allele?"	A gene is the instruction for a trait, like eye color. An allele is one particular version of it, like the brown version or the blue version.
"If it is 3 to 1, does that mean out of four children exactly one will have it?"	No. It is a 1 in 4 chance for each child separately, like flipping a coin. A family of four could easily have none or all four.
"Why do capital and lowercase letters matter?"	Capital means the dominant version, lowercase means the recessive version. Always the same letter, just different case.

FINISHED EARLY?

The Go Further page asks about a family with four children where the odds did not come out as predicted, which is where the probability idea actually lands.

Read: Two Copies of Everything

- 1 Every one of your cells carries two copies of nearly every gene, one inherited from each parent. A gene is the instruction for a trait. The different versions of that instruction are called **alleles**.
- 2 When the two copies disagree, one version often masks the other. The one that shows is called **dominant** and is written as a capital letter. The one that is hidden is **recessive** and is written as the same letter in lowercase.
- 3 Take a plant where round seeds (**R**) are dominant and wrinkled seeds (**r**) are recessive. A plant with **RR** has round seeds. A plant with **rr** has wrinkled seeds. A plant with **Rr** has round seeds too, because one dominant allele is enough to show. That last plant is a carrier: it looks round but is quietly carrying a wrinkled allele it can pass on.
- 4 Here is the useful part. Each parent passes on only **one** of their two alleles, chosen at random. A Punnett square is a grid that lays out every combination that could result.

	R	r
R		
r		

Father across the top, mother down the side

WORKED EXAMPLE, DONE FOR YOU

Both parents are **Rr**. The father's two alleles go across the top, the mother's down the side. Fill each box by combining the letter above it with the letter beside it.

Top left is **RR**. Top right is **Rr**. Bottom left is **Rr**. Bottom right is **rr**.

So the four possible outcomes are **RR, Rr, Rr, rr**. Three of the four have at least one capital R, so three out of four will have round seeds and one out of four will be wrinkled. That is the famous **3 to 1** ratio.

THE PART ALMOST EVERYONE GETS WRONG

That 3 to 1 is a **probability for each offspring separately**, not a promise about a group. It is like flipping a coin: a 1 in 4 chance each time. Four seeds could easily come out four round and none wrinkled.

Questions: Where Traits Come From

In this species, **B** gives black fur and is dominant. **b** gives white fur and is recessive.

1 Fill in this square. Both parents are Bb.

	B	b	RESULTS
B			Black: ____ of 4
b			White: ____ of 4

2 Fill in this square. One parent is BB, the other is bb.

	B	b	RESULTS
B			Black: ____ of 4
b			White: ____ of 4

3 Fill in this square. One parent is Bb, the other is bb.

	B	b	RESULTS
B			Black: ____ of 4
b			White: ____ of 4

4 Two black animals are bred together and one of their offspring is white. What must the alleles of both parents have been? Explain how you know.

5 A white animal is bred with a black animal and every single offspring is black. What can you say about the black parent?

Go Further: Four Children

THE SITUATION

In humans, the ability to roll your tongue into a tube is controlled by a single gene in this simplified model. Rolling (**T**) is dominant. Not rolling (**t**) is recessive.

Two parents can both roll their tongues. They have four children. **Three of the four cannot roll their tongues.**

1 What must both parents' alleles be? Explain your reasoning.

2 Draw the Punnett square for these parents and write the predicted ratio.

	----	----	PREDICTED RATIO

3 The prediction and the real family do not match. Does that mean the prediction was wrong? Explain carefully.

CHALLENGE

A breeder wants to know whether a black animal is **BB** or **Bb**, but cannot tell by looking because both are black. Which animal should they breed it with to find out, and what result would prove it was **Bb**? Explain your test.

Answer Key: Where Traits Come From

QUESTION PAGE

- Boxes: BB, Bb, Bb, bb. **Black 3 of 4, white 1 of 4.**
- Every box is Bb. **Black 4 of 4, white 0 of 4.** All offspring are black but every one carries a hidden white allele.
- Boxes: Bb, bb, Bb, bb. **Black 2 of 4, white 2 of 4,** a 1 to 1 ratio.
- Both parents must be **Bb**. A white offspring is bb, so it received a b from each parent. Since both parents look black, each must have had one B and one b.
This is the strongest question on the page for showing real understanding. It runs the logic backward from offspring to parent.
- The black parent is almost certainly **BB**. If it were Bb, roughly half the offspring would be expected to be white.
Accept "probably BB" and reward any student who notes that with a small number of offspring you cannot be completely certain. That is excellent scientific thinking.

GO FURTHER PAGE

- Both parents must be **Tt**. They can roll, so each has at least one T. Their non-rolling children are tt, so each parent also carried a t.
- Square gives TT, Tt, Tt, tt. Predicted ratio **3 rollers to 1 non-roller.**
- No.** The prediction gives each child separately a 1 in 4 chance of being a non-roller. It never promised that exactly one of four would be. Three non-rollers out of four is unlikely but perfectly possible, in the same way four coin flips can land three tails.
This is the point of the whole lesson. Full credit requires the idea that the ratio is a per-child probability, not a quota. Common wrong answer: "the parents must have different alleles than we thought." Ask them what the odds of three tails in four flips are.

Challenge: Breed it with a **white (bb)** animal. If any offspring is white, the unknown parent must have carried a b and is therefore Bb. If a large number of offspring are all black, it is very likely BB.

This is a real technique, called a test cross. Reward any student who notes that one all-black litter is suggestive rather than conclusive.

MARKING SHORTCUT

If you are short on time, mark question 4 and Go Further question 3 only. Between them they show whether the student can run the logic backward from offspring to parent, and whether they understand that a ratio is a probability rather than a quota. Everything else on the pages is practice toward those two ideas.

IF A STUDENT FINISHES EVERYTHING

Ask them to work out the Punnett square for two parents who are both **BB**, then explain why breeders sometimes deliberately keep an animal that carries a recessive allele rather than breeding it out.

The Beetles That Survived

TEACHER BRIEFING • KEEP THIS PAGE

NO SCIENCE BACKGROUND NEEDED. NO LAB, NO EQUIPMENT.

Everything you need is on this page and the answer key. Nothing in this lesson is heated, poured, cut or plugged in.

Copy for students: the next three pages. **Keep:** this page and the answer key.

What this lesson does

Students work through four generations of a beetle population using real numbers, and watch a trait become common without any individual beetle changing. The lesson exists to kill the single most persistent misconception in biology: that organisms change themselves in response to their environment.

How the period runs

MIN	WHAT YOU DO
0-5	Read the opening lines below. Hand out all three student pages.
5-14	Students read the scenario and the generation table.
14-40	Question page. Question 5 is the one that matters.
40-53	Go Further: the antibiotic version of the same story.
53-55	Collect all pages.

SAY THIS TO START

I want you to watch out for one word today, and the word is tried. No beetle in this story tried to become green. No beetle wanted to change color, and no beetle changed color during its life. Every beetle was born whatever color it was born and stayed that color until it died. And yet after four generations, almost the whole population is green. Your job is to work out how that happened.

The three questions students will ask

THEY ASK	YOU SAY
"Did the beetles change color to hide?"	No. Each beetle keeps the color it was born with. The population changed because green ones survived more often and had more offspring.
"Where did the first green beetle come from?"	A random mutation, which is a copying error in DNA. Most mutations do nothing or are harmful. Occasionally one happens to be useful.
"Is evolution just a theory?"	In science, a theory means an explanation supported by a very large body of evidence, not a guess. Say that plainly and move on.

FINISHED EARLY?

The Go Further page applies the identical logic to bacteria and antibiotics, which is the version students are most likely to meet outside school.

Read: Four Generations

Four Generations

A worked scenario • written for this lesson

- 1 On an island there lives a beetle that comes in two colors. Most are brown. A very small number are green. The color a beetle has is inherited from its parents and never changes during its life. A brown beetle stays brown. A green beetle stays green.
- 2 For a long time the island's plants were brown and dry, and the brown beetles were nearly invisible against them. The green ones stood out, got eaten by birds more often, and stayed rare.
- 3 Then the island's climate became wetter. Over a few years the vegetation turned green and stayed green. Nothing about the beetles changed. What changed was the background they sat on.
- 4 Scientists counted the beetles once per generation for four generations after the change.

GENERATION	BROWN BEETLES	GREEN BEETLES	TOTAL	PERCENT GREEN
1 (start)	960	40	1000	4%
2	720	180	900	20%
3	390	560	950	59%
4	130	840	970	87%

- 5 Notice what the numbers do and do not say. The total population stayed roughly the same throughout, close to a thousand beetles. No beetle in generation 2 was a recolored version of a beetle from generation 1, because beetles do not change color. Every beetle in generation 2 was the offspring of beetles from generation 1.
- 6 What actually happened is simpler and stranger than it looks. On green vegetation, green beetles were harder for birds to spot. Slightly more of them survived long enough to reproduce. Because color is inherited, their offspring were green too. Repeat that over four generations and a trait that was once rare becomes the normal one.
- 7 This process has three requirements, and all three must be present or nothing happens. First, there must be **variation**: the beetles were not all identical to begin with. Second, the variation must be **inherited**: color passes from parent to offspring. Third, the variation must **affect survival or reproduction** in that particular environment. Take away any one of the three and the population would not have shifted at all.
- 8 It is worth being precise about the word *adaptation*. A population becomes adapted. An individual does not adapt. The green beetle was not better suited to brown vegetation and then improved. It was always green, and the environment moved to meet it.

Questions: The Beetles That Survived

1 Between generation 1 and generation 4, how many more green beetles were there? Show your working.

2 The percent of green beetles rose from 4% to 87%. Did the number of brown beetles go up, go down, or stay the same? Give the figures.

3 Paragraph 3 says nothing about the beetles changed. What changed instead, and why did that matter to the beetles?

4 List the three requirements from paragraph 7 in your own words.

5 A student writes: "*The beetles turned green so the birds could not see them.*" This sentence is wrong in two separate ways. Explain both.

This is the most important question on the page.

6 Suppose the island dried out again and the plants turned brown. Predict what would happen to the population over the next few generations, and explain why.

THE SITUATION

A hospital treats an infection with an antibiotic. In any large population of bacteria, a very small number happen to carry a random mutation that makes the antibiotic less effective against them. The antibiotic is used widely for two years.

TIME	KILLED BY DRUG	SURVIVE DRUG	PERCENT SURVIVING
Before treatment starts	999,000	1,000	0.1%
After 6 months of use	94,000	6,000	6%
After 2 years of use	42,000	58,000	58%

1 Match each part of this story to the beetle story. Fill the table.

THE THREE REQUIREMENTS	IN THE BEETLES	IN THE BACTERIA
Variation exists		
It is inherited		
It affects survival		

2 Did the antibiotic cause the mutation that let some bacteria survive? Explain carefully.

CHALLENGE

Doctors tell patients to finish the whole course of antibiotics even after they start feeling better. Using what you have just worked out, explain why stopping early could make the problem worse.

QUESTION PAGE

1. 840 minus 40 equals **800 more green beetles**.
2. **Down sharply**, from 960 to 130. Worth noticing that the total population barely changed, so this is a change in proportions, not a population crash.
3. The vegetation turned green. That mattered because it changed which color was camouflaged, so green beetles became harder for birds to spot and brown ones became easier.
4. 1. The beetles were not all the same to start with. 2. Color is passed from parent to offspring. 3. In that environment, color affected the chance of surviving long enough to reproduce.
5. **Error one:** no beetle turned green. Each beetle kept the color it was born with. The *population* changed, not any individual. **Error two:** the word "so" implies purpose, as if the beetles chose it or aimed at it. Nothing here is aimed at anything. Green beetles simply survived more often, and survival is not a decision.
This is the question to mark carefully. Most students get error one and miss error two. Error two is the deeper one: the language of purpose is what makes evolution feel like it has a plan. If they only get one, half credit and write the other one in.
6. The proportions would reverse over several generations. Brown would once again be camouflaged and green would stand out, so brown beetles would survive more often and become common again. Provided some brown beetles remain, which they do at 130.
Strong students notice that if brown had reached zero, it could not come back without a new mutation. Excellent point, give extra credit.

GO FURTHER PAGE

REQUIREMENT	BEETLES	BACTERIA
Variation	Some brown, some green	A small number carry a resistance mutation
Inherited	Color passes to offspring	The mutation passes to daughter cells
Affects survival	Camouflage against birds	Surviving the antibiotic

2. **No.** The mutations were already present by chance in a tiny fraction of the population before the drug arrived. The antibiotic did not create them. It removed everything else, which left the resistant ones to multiply.
This is the exact parallel to question 5 and it is the payoff of the lesson. Common wrong answer: "the antibiotic made them stronger."
- Challenge:** Stopping early kills the most susceptible bacteria first and leaves behind the hardest to kill. Those survivors then multiply, so the remaining population is more resistant than the one you started with.

Trading Gases

TEACHER BRIEFING • KEEP THIS PAGE

NO SCIENCE BACKGROUND NEEDED. NO LAB, NO EQUIPMENT.

Everything you need is on this page and the answer key. Nothing in this lesson is heated, poured, cut or plugged in.

Copy for students: the next three pages. **Keep:** this page and the answer key.

What this lesson does

Students trace where a tree's mass actually comes from, then map the two processes as a loop rather than as two memorized word equations. The lesson targets two misconceptions at once: that plants get their mass from soil, and that plants do not respire.

How the period runs

MIN	WHAT YOU DO
0-5	Read the opening lines below. Hand out all three student pages.
5-13	Students read the page.
13-37	Question page. The loop diagram in question 4 is the core task.
37-48	Go Further: the sealed terrarium.
48-50	Collect all pages.

SAY THIS TO START

Here is a question that sounds easy and is not. A tree seed weighs less than a paperclip. Thirty years later that tree weighs two tons. Where did the two tons come from? Almost everyone says soil, and almost everyone is wrong. If it came from soil there would be a two-ton hole under every big tree, and there is not.

The three questions students will ask

THEY ASK	YOU SAY
"Do plants breathe?"	Plants respire constantly, day and night, just like animals. They also photosynthesize, but only in light. Respiration never stops.
"So where does the mass come from?"	Mostly carbon dioxide from the air, plus water. The carbon in the wood was floating past as gas before the tree caught it.
"Why do plants need soil then?"	For water, for minerals in small amounts, and to hold them upright. Not for the bulk of their mass.

FINISHED EARLY?

The Go Further page asks what happens inside a sealed glass terrarium over 24 hours, which forces students to use both processes together.

Read: Where the Tree Came From

- 1 Around 1640 a Flemish scientist named Jan Baptist van Helmont ran an experiment that took him five years. He weighed a pot of dry soil, planted a small willow shoot in it, weighed that too, and then did nothing but add rainwater.
- 2 After five years he pulled the tree out and weighed everything again. The willow had gained about 164 pounds. The soil had lost about 2 ounces. Van Helmont concluded that the tree was made of water, since water was the only thing he had added. He was mostly wrong, but the experiment was excellent, and the part he proved is the part that still matters: **the mass did not come from the soil.**
- 3 It came from the air. Specifically it came from carbon dioxide, a gas that makes up a very small fraction of the atmosphere and that plants pull out of it continuously. Inside the leaves, using energy captured from sunlight, plants combine carbon dioxide and water to build sugar. Oxygen is left over and released. That process is **photosynthesis**.
- 4 Written out: carbon dioxide plus water, using light energy, becomes sugar plus oxygen. The sugar is the point. It is the plant's food and the raw material for wood, leaves, roots and fruit. The carbon locked in a wooden table was drifting past as gas before a tree caught it.
- 5 Making sugar is only half of it. Having made food, the plant then has to get the energy back out of it, and it does this exactly the way you do. **Cellular respiration** takes sugar and oxygen and releases energy, giving off carbon dioxide and water as waste. Every living cell does this, in plants and animals alike, continuously.
- 6 Look carefully at those two and you will see something neat. The inputs of one are the outputs of the other. "Photosynthesis takes in carbon dioxide and water and gives out sugar and oxygen. Respiration takes in sugar and oxygen and gives out carbon dioxide and water. They run in opposite directions.
- 7 This is where the common mistake lives. People often say plants photosynthesize and animals respire, as if it were a division of labor. It is not. **Plants do both.** A plant respire twenty four hours a day, in the dark as much as in the light. It only photosynthesizes when there is light.
- 8 So during a bright day a plant photosynthesizes faster than it respire, and it releases oxygen overall. At night photosynthesis stops entirely while respiration carries on, so the same plant takes in oxygen and gives out carbon dioxide, exactly like you do. The plant on your windowsill is not doing one job. It is doing two, and only one of them clocks off at sunset.

Questions: Trading Gases

1 In van Helmont's experiment, how much mass did the tree gain and how much did the soil lose? What does that comparison prove?

2 Van Helmont concluded the tree was made of water. What was he missing?

3 Complete the table for both processes.

	GOES IN	COMES OUT	WHEN IT HAPPENS
Photosynthesis			
Cellular respiration			

4 Explain in your own words why the two processes are described as running in opposite directions.

5 True or false: *plants photosynthesize and animals respire*. Explain your answer.

6 A sealed greenhouse is measured at 2 p.m. and again at 2 a.m. At which time would the oxygen level be higher? Explain.

Go Further: The Sealed Terrarium

THE SITUATION

A sealed glass jar contains soil, a small healthy plant, and air. Nothing can get in or out. It sits on a windowsill. Gas levels inside are measured every six hours for one full day.

TIME	LIGHT?	OXYGEN LEVEL	CARBON DIOXIDE LEVEL
6 a.m.	Dim	20.6%	0.9%
12 noon	Bright	21.4%	0.4%
6 p.m.	Dim	21.5%	0.3%
12 midnight	Dark	20.9%	0.7%

1 Between 6 a.m. and 12 noon, which process was running faster? How do the numbers tell you?

2 Between 6 p.m. and midnight, oxygen fell and carbon dioxide rose. Explain exactly what was happening inside the jar during those hours.

3 Notice that oxygen never reaches zero and carbon dioxide never runs out. Explain why the jar can keep going.

CHALLENGE

A student says the plant would survive longer if you put the jar in a completely dark cupboard, because then it would not have to work so hard. Explain what is wrong with this and predict what would actually happen.

Answer Key: Trading Gases

QUESTION PAGE

1. The tree gained about 164 pounds. The soil lost about 2 ounces. Since the soil lost almost nothing, the tree's mass cannot have come from it.
2. Carbon dioxide from the air. He never thought to weigh the air, because gases were not understood as having mass at the time.

	IN	OUT	WHEN
Photosynthesis	Carbon dioxide + water (+ light energy)	Sugar + oxygen	Only in light
Respiration	Sugar + oxygen	Carbon dioxide + water (+ energy released)	All the time, day and night

4. What goes into one comes out of the other. Photosynthesis takes carbon dioxide and water and produces sugar and oxygen; respiration takes sugar and oxygen and produces carbon dioxide and water.
5. **False.** Plants do both. Plants respire continuously, day and night, and photosynthesize only in light. Animals respire and do not photosynthesize.
The most common wrong answer is "true" with a confident explanation. This is worth correcting on the board if you have time.
6. **2 p.m.** In daylight photosynthesis outpaces respiration, so oxygen builds up. At 2 a.m. photosynthesis has stopped but respiration continues, so oxygen is being used up.

GO FURTHER PAGE

1. Photosynthesis. Oxygen rose from 20.6% to 21.4% and carbon dioxide fell from 0.9% to 0.4%, which is the signature of photosynthesis running faster than respiration.
2. The light faded, so photosynthesis slowed and then stopped. Respiration carried on all the way through, consuming oxygen and producing carbon dioxide, so the levels moved back the other way.
3. Because the two processes recycle each other's waste. The oxygen respiration uses is replaced by photosynthesis in daylight, and the carbon dioxide photosynthesis uses is replaced by respiration. It is a loop, not a one-way consumption.

Challenge: Wrong. In the dark the plant cannot photosynthesize at all, but respiration continues regardless, so it would steadily consume oxygen and its own stored sugar with no way to replace either. It would die faster in the dark, not slower.
Look for the recognition that respiration does not stop. That is the whole lesson in one answer.

The Delivery Network

TEACHER BRIEFING • KEEP THIS PAGE

NO SCIENCE BACKGROUND NEEDED. NO LAB, NO EQUIPMENT.

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Copy for students: the next three pages. **Keep:** this page and the answer key.

What this lesson does

Students trace a single oxygen molecule and a single glucose molecule from outside the body to a working muscle cell, and back out again as waste. Body systems are usually taught as separate labeled diagrams; this lesson teaches them as one connected delivery problem, which is what they actually are.

How the period runs

MIN	WHAT YOU DO
0-5	Read the opening lines below. Hand out all three student pages.
5-13	Students read the page.
13-38	Question page. The ordering task in question 2 is the core.
38-48	Go Further: what happens when you sprint.
48-50	Collect all pages.

SAY THIS TO START

Your body has about thirty trillion cells and every single one of them needs two deliveries, constantly, or it dies within minutes. It needs oxygen and it needs fuel. No cell is more than a fraction of a millimeter from a blood vessel, because that is how tight the delivery schedule is. Today you are going to follow the delivery route from the outside world all the way to one muscle cell in your leg.

The three questions students will ask

THEY ASK	YOU SAY
"Why does blood need to go back to the heart between trips?"	Because it makes two separate journeys. One short trip to the lungs to collect oxygen, then back to the heart to be pumped out to the body at higher pressure.
"Is blue blood real?"	No. Blood low in oxygen is dark red, not blue. Veins look bluish through skin because of how light passes through tissue.
"Why do you breathe harder when you exercise?"	Muscles are respiring faster, so they need oxygen delivered faster and carbon dioxide removed faster. Breathing rate and heart rate both rise to keep up.

FINISHED EARLY?

The Go Further page walks through what changes in both systems during a sprint, and why you keep breathing hard after you stop.

Read: One Delivery Route

- 1 Every cell in your body runs cellular respiration: it takes in glucose and oxygen, releases energy, and produces carbon dioxide and water as waste. That means every cell needs two things delivered and one thing taken away, without interruption, for your whole life.
- 2 Two organ systems handle it together. The **respiratory system** gets oxygen into the body and carbon dioxide out. The **circulatory system** carries both of them, plus fuel, to and from every cell. Neither works without the other.
- 3 Start with a breath. Air travels down the trachea, splits into two bronchi, and branches into narrower and narrower tubes until it reaches roughly 300 million tiny sacs called **alveoli**. These are the point of the whole system. Their walls are one cell thick, and they are wrapped in blood vessels that are also one cell thick.
- 4 That thinness is the design. Oxygen crosses from the air in the alveolus into the blood, and carbon dioxide crosses the other way, simply by moving from where there is more of it to where there is less. Unfolded and laid flat, the alveoli of one adult would cover roughly half a tennis court, all packed inside your chest.
- 5 Now the blood carries the oxygen, held by **hemoglobin** inside red blood cells. It returns to the heart, which pumps it out through arteries. Arteries branch into smaller vessels and finally into **capillaries**, which are so narrow that red blood cells pass through in single file. This is where delivery actually happens. Oxygen and glucose leave the blood, carbon dioxide enters it.
- 6 The blood, now low in oxygen and carrying carbon dioxide, flows into veins and returns to the heart. From there it is pumped a short distance to the lungs, where carbon dioxide crosses into the alveoli and is breathed out, and fresh oxygen is loaded. Then back to the heart again to be sent around the body.
- 7 That is why the heart has four chambers. Blood makes two loops, not one: a short loop to the lungs and a long loop to the body, and the heart is really two pumps side by side keeping the oxygen-rich and oxygen-poor blood separate.
- 8 The glucose comes from somewhere else entirely. Food is broken down in the **digestive system** and glucose is absorbed through the wall of the small intestine into the blood. So a muscle cell in your leg is being served by three systems at once: digestive for fuel, respiratory for oxygen, circulatory for the delivery run.

Questions: The Delivery Network

1 What two things does every cell need delivered, and what one thing must be taken away?

2 Put these eight steps in the correct order, numbering them 1 to 8. Follow one oxygen molecule from the air to a muscle cell in the leg.

ORDER	STEP
	Oxygen crosses into a capillary in the leg and enters the muscle cell
	Air is drawn down the trachea
	Blood is pumped out of the heart into arteries
	Oxygen crosses the alveolus wall into the blood
	Oxygen attaches to hemoglobin in a red blood cell
	Air reaches the alveoli
	Arteries branch into capillaries
	Oxygen-rich blood returns to the heart

3 Alveoli walls are one cell thick and so are the blood vessels around them. Why does that thinness matter?

4 Why does the heart have four chambers rather than two? Answer using the idea of two loops.

5 A muscle cell needs glucose as well as oxygen. Which system supplies it, and how does it reach the blood?

Go Further: What Changes When You Sprint

THE DATA

Measurements taken from one person at rest, during a 200 meter sprint, and two minutes after finishing.

MEASUREMENT	AT REST	SPRINTING	2 MIN AFTER
Heart rate (beats/min)	68	186	124
Breaths per minute	14	42	29
Blood to leg muscles	Low	Very high	High

1 Explain why both heart rate and breathing rate rise during the sprint. Give a separate reason for each.

2 The sprint is over, yet two minutes later both rates are still well above resting. Suggest why the body does not simply return to normal immediately.

3 During the sprint, blood flow to the leg muscles goes up sharply. Where do you think that extra blood is diverted from, and why is that a sensible trade?

CHALLENGE

Athletes sometimes train at high altitude, where there is less oxygen in each breath. Over several weeks their bodies produce more red blood cells. Using what you know about hemoglobin, explain why that would help them when they return to sea level.

Answer Key: The Delivery Network

QUESTION PAGE

1. Delivered: oxygen and glucose. Removed: carbon dioxide. Accept water as an additional waste product.
2. **1** Air drawn down the trachea. **2** Air reaches the alveoli. **3** Oxygen crosses the alveolus wall into the blood. **4** Oxygen attaches to hemoglobin. **5** Oxygen-rich blood returns to the heart. **6** Blood pumped out into arteries. **7** Arteries branch into capillaries. **8** Oxygen crosses into the muscle cell.
The step students most often misplace is number 5. Many assume the lungs send blood straight to the body. Worth pointing out that everything goes back through the heart first.
3. The thinner the barrier, the faster gases can cross it. A thick wall would slow oxygen entering the blood and carbon dioxide leaving, and the delivery schedule would fail.
4. Blood makes two separate loops: a short one to the lungs and a long one to the body. Four chambers keep oxygen-rich and oxygen-poor blood from mixing, so the heart works as two pumps side by side.
5. The digestive system. Food is broken down and glucose is absorbed through the wall of the small intestine into the blood.

GO FURTHER PAGE

1. **Breathing rate** rises to load oxygen into the blood faster and unload carbon dioxide faster. **Heart rate** rises to move that blood to the muscles faster. One is about getting gases in and out; the other is about delivery speed.
Full credit needs two distinct reasons. "To get more oxygen" twice is half credit.
 2. The muscles still have carbon dioxide to clear and an oxygen shortfall to repay, so the systems keep running fast until the backlog is cleared.
Do not require the term oxygen debt. The idea of a backlog is the target.
 3. From the digestive system and other organs not urgently needed during a sprint. Sensible because digestion can wait a few minutes while muscles cannot.
- Challenge:** More red blood cells means more hemoglobin, so each unit of blood carries more oxygen. Back at sea level, where oxygen is plentiful, they can deliver more oxygen per heartbeat than before.

Why It Floats

TEACHER BRIEFING · KEEP THIS PAGE

NO SCIENCE BACKGROUND NEEDED. NO LAB, NO EQUIPMENT.

Everything you need is on this page and the answer key. Nothing in this lesson is heated, poured, cut or plugged in.

Copy for students: the next three pages. **Keep:** this page and the answer key.

What this lesson does

Students calculate density for six objects and use the results to predict floating. The lesson attacks the belief that heavy things sink, which survives most teaching because it is right often enough to feel true. A 100,000 ton ship and a 2 gram paperclip settle it.

How the period runs

MIN	WHAT YOU DO
0-6	Read the opening lines below. Hand out all three student pages. Work through the first row of the table with the class.
6-13	Students read the page.
13-40	Question page. Six density calculations, then predictions.
40-53	Go Further: the steel ship problem.
53-55	Collect all pages.

SAY THIS TO START

Two objects. A steel paperclip weighing two grams, and a container ship weighing a hundred thousand tons. Drop them both in the ocean. The paperclip sinks and the ship floats. So whatever decides floating, it is not weight. Today you will work out what it actually is, and you can do it with division.

The three questions students will ask

THEY ASK	YOU SAY
"What is the formula?"	Density = mass divided by volume. It is printed on the student page. Units here are grams per cubic centimeter.
"What number do we compare to?"	Water is 1.0 g/cm ³ . Less than 1.0 floats, more than 1.0 sinks. That is the whole rule.
"Why does a steel ship float then?"	Because a ship is mostly air inside. Take the whole ship including the air and divide by its total volume and the answer is less than 1.0.

FINISHED EARLY?

The Go Further page has students calculate the density of a hollow steel hull and compare it to a solid steel block of the same mass.

Read: Heavy Is Not the Same as Dense

THE FORMULA YOU NEED

$$\text{density} = \text{mass} \div \text{volume}$$

Mass in grams, volume in cubic centimeters (cm^3), density in g/cm^3 .

Water has a density of $1.0 \text{ g}/\text{cm}^3$.

- 1 A kilogram of feathers and a kilogram of lead weigh exactly the same, which everybody knows, and yet the feathers fill a room while the lead sits in your palm. The difference is not mass. It is how much space that mass takes up. That is **density**: how much stuff is packed into a given amount of space.
- 2 To find it you divide the mass by the volume. A block with a mass of 60 grams occupying 20 cubic centimeters has a density of $3.0 \text{ g}/\text{cm}^3$. Every cubic centimeter of it holds 3 grams, wherever you cut it. Density is a property of the material, not of the size of the lump. Cut that block in half and each half still has a density of 3.0.
- 3 Here is what makes it useful. Whether something floats in water does not depend on its mass at all. It depends on whether its density is greater or less than the density of water. Water is $1.0 \text{ g}/\text{cm}^3$. Anything below that floats. Anything above it sinks. A two gram paperclip sinks because steel is about $7.8 \text{ g}/\text{cm}^3$. A one ton log floats because wood is about 0.5.
- 4 The reason is a push. When an object sits in water it pushes some water out of the way, and the water pushes back upward. This upward push is called **buoyancy**, and its size equals the weight of the water that got displaced. If the water pushed aside weighs more than the object, the object floats. If it weighs less, the object sinks. Which is exactly the same statement as the density rule, said a different way.
- 5 Now the ship. Steel really is $7.8 \text{ g}/\text{cm}^3$, and a solid steel block really does sink. But a ship is not a solid steel block. It is a thin steel shell wrapped around an enormous volume of air. When you calculate the density of the **whole ship**, including all that air, you divide a very large mass by an even larger volume, and the answer comes out below 1.0.
- 6 This is why shape matters so much and mass matters so little. Take a lump of modeling clay and it sinks. Take the exact same lump, with the exact same mass, and press it into a bowl shape, and it floats. Nothing was added or removed. The volume went up because the bowl encloses air, so the density went down, so it floats.
- 7 It also explains a hole in a hull. Water entering a ship replaces air with something 800 times denser. Total mass rises while total volume stays the same, so the average density climbs. Once it passes 1.0, the ship goes down. Nothing about the steel changed.

Questions: Why It Floats

Calculate the density of each object, then predict whether it floats or sinks in water. Show your division.

OBJECT	MASS (G)	VOLUME (CM3)	DENSITY (G/CM3)	FLOAT OR SINK?
Oak block	180	240		
Aluminum cube	135	50		
Candle wax	90	100		
Glass marble	25	10		
Cork stopper	6	25		
Ice cube	92	100		

1 The oak block has a mass of 180 g. The glass marble has a mass of only 25 g. Yet the marble sinks and the block floats. Explain why mass is the wrong thing to look at.

2 Ice is made of water, yet its density is not 1.0. What does your answer tell you about why ice floats on water?

3 A student cuts the oak block exactly in half. What is the density of each half? Explain.

Go Further: Building a Ship

THE PROBLEM

You have **7,800 grams of steel**. Steel has a density of 7.8 g/cm^3 , so that steel occupies $1,000 \text{ cm}^3$ when solid. You are going to use the same steel two different ways.

VERSION	MASS (G)	TOTAL VOLUME (CM3)	DENSITY	FLOATS?
A. Solid steel block	7,800	1,000		
B. Same steel beaten into a hull enclosing air	7,800	12,000		

1 Both versions contain exactly the same steel and have exactly the same mass. Explain in your own words why one floats and the other does not.

2 Version B springs a leak and takes on 6,000 g of water, which adds mass but no extra volume. Calculate the new density and state what happens.

CHALLENGE

A submarine can choose to float or sink without changing shape. It has tanks that can be filled with either air or seawater. Explain how that lets it control its density, and what the crew would do to make it rise.

IF YOU STILL HAVE TIME

Saltwater has a density of about 1.03 g/cm^3 . Would a ship float higher or lower in the ocean than in a freshwater lake? Explain.

Answer Key: Why It Floats

QUESTION PAGE

OBJECT	WORKING	DENSITY	RESULT
Oak block	180 / 240	0.75	Floats
Aluminum cube	135 / 50	2.7	Sinks
Candle wax	90 / 100	0.9	Floats
Glass marble	25 / 10	2.5	Sinks
Cork stopper	6 / 25	0.24	Floats
Ice cube	92 / 100	0.92	Floats

1. Because floating depends on density, not mass. The oak block is heavier but spreads its mass over far more volume, giving 0.75. The marble packs less mass into much less space, giving 2.5.

2. Ice is 0.92, less than liquid water at 1.0, so it floats. Water is unusual: it expands when it freezes, so the same mass takes up more space as ice.

Worth mentioning if there is time: almost every other substance is denser as a solid. If ice sank, lakes would freeze from the bottom up.

3. Still **0.75**. Both mass and volume halve, so the division gives the same answer. Density is a property of the material, not the size of the piece.

A very common wrong answer is 0.375. If several students give it, do the arithmetic on the board: 90 / 120.

GO FURTHER PAGE

VERSION	WORKING	DENSITY	FLOATS?
A. Solid block	7,800 / 1,000	7.8	No, sinks
B. Hull	7,800 / 12,000	0.65	Yes, floats

1. The mass is identical but the hull encloses a large volume of air, so the total volume is twelve times greater. Dividing the same mass by a much larger volume gives a much smaller density, and 0.65 is below water's 1.0.

2. New mass 13,800 g, volume still 12,000 cm³. $13,800 / 12,000 = 1.15$. That is above 1.0, so it sinks.

Challenge: Filling the tanks with seawater adds mass without changing the submarine's volume, raising its density above 1.0 so it sinks. To rise, the crew blows compressed air into the tanks to force the water out, lowering the mass and therefore the density.

Saltwater: **Higher.** Saltwater is denser, so a smaller volume of it has to be displaced to support the same ship.

Closing the Loop

TEACHER BRIEFING · KEEP THIS PAGE

NO SCIENCE BACKGROUND NEEDED. NO LAB, NO EQUIPMENT.

Everything you need is on this page and the answer key. Nothing in this lesson is heated, poured, cut or plugged in.

Copy for students: the next three pages. **Keep:** this page and the answer key.

What this lesson does

Students read two circuit diagrams and predict what happens when a bulb is removed from each. The lesson is built around one question with real consequences: why does one dead bulb kill a whole string of lights while one dead lamp in a house changes nothing?

How the period runs

MIN	WHAT YOU DO
0-5	Read the opening lines below. Hand out all three student pages.
5-13	Students read the page and study the two circuits.
13-37	Question page.
37-48	Go Further: wiring a house.
48-50	Collect all pages. Nothing here is plugged in or switched on.

SAY THIS TO START

Everything about circuits comes back to one word: loop. Electricity needs a complete path from one end of the battery all the way back around to the other. Break the loop anywhere and everything stops, everywhere. That single idea explains switches, fuses, dead bulbs and why your bedroom light does not go off when the kitchen one does.

The three questions students will ask

THEY ASK	YOU SAY
"Is this dangerous, can we test it?"	No equipment today. Everything on the page is a diagram. Nothing is plugged in and nothing needs to be.
"Does electricity get used up as it goes round?"	No. The current is the same everywhere in a series loop. What gets used up is energy, which the components convert to light and heat.
"Which is better, series or parallel?"	Neither. They do different jobs. Parallel is right for house wiring; series is right when you want one switch to control everything.

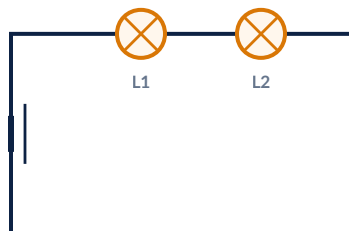
FINISHED EARLY?

The Go Further page asks students to wire an imaginary room and justify which parts should be series and which parallel.

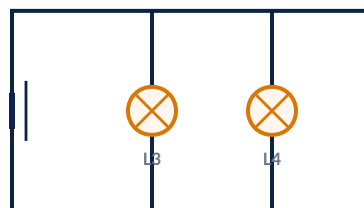
Read: Two Ways to Wire It

STUDY THESE TWO CIRCUITS

Both have the same battery and two identical lamps. The only difference is how the lamps are connected.



Circuit A



Circuit B

- 1 An electric circuit is a complete loop. A source of energy, usually a battery, pushes charge around a path of conducting material and back to where it started. If the loop is complete, current flows. If the loop is broken at any single point, current stops everywhere in that loop, instantly. That is all a switch does: it makes a gap in the loop and closes it again.
- 2 **Circuit A is a series circuit.** There is exactly one path. The current leaves the battery, passes through the first lamp, then through the second lamp, then back to the battery. Every bit of charge must go through both lamps because there is nowhere else for it to go.
- 3 This creates a weakness that anyone who has untangled old holiday lights already knows. Remove one lamp from a series circuit, or let one burn out, and you have made a gap in the only path. Everything goes dark, including the lamps that are still perfectly good.
- 4 Series has another effect. The two lamps share the battery's push between them, so each one gets less than it would on its own, and both glow dimmer than a single lamp would. Add a third and all three get dimmer still.
- 5 **Circuit B is a parallel circuit.** Here the path splits. Current leaving the battery arrives at a junction and divides, with some going through one lamp and some through the other, before rejoining and returning to the battery. Each lamp sits on its own branch.
- 6 Now remove one lamp. You have broken one branch, but the other branch is still a complete loop, so the remaining lamp carries on exactly as before. It does not even dim, because each branch gets the full push of the battery rather than a share of it.
- 7 This is why every building you have ever been in is wired in parallel. If houses were wired in series, switching off a bedside lamp would black out the entire street. It also means each appliance gets the full supply rather than a fraction that shrinks every time somebody plugs in a toaster.
- 8 Parallel has a cost. Because each new branch is an extra path, adding branches lets more total current flow from the source. Overload a circuit with too many branches and the wires heat up. That is what a fuse or a circuit breaker is for: a deliberately weak point that breaks the loop before anything else does.

Questions: Closing the Loop

1 In your own words, what has to be true before any current will flow?

2 Predict what happens in each case. Give the reason each time.

WHAT YOU DO	WHAT HAPPENS	WHY
Remove lamp L1 from circuit A		
Remove lamp L3 from circuit B		
Add a switch anywhere in circuit A and open it		
Add a third lamp in series to circuit A		

3 Which circuit gives brighter lamps, A or B? Explain using the idea of the battery's push being shared or not shared.

4 Explain why an old string of holiday lights goes completely dark when one bulb burns out, but your kitchen light staying on does not depend on the bathroom light.

5 Paragraph 8 says a fuse is a deliberately weak point. Explain why building in a weak point on purpose is a good idea.

Go Further: Wiring a Room

THE JOB

You are wiring one room. It contains a ceiling light, a desk lamp, a heater, and a single main switch by the door that must be able to cut power to the whole room at once. Decide how each part should be connected.

PART	SERIES OR PARALLEL?	WHY YOU CHOSE IT
Ceiling light		
Desk lamp		
Heater		
Main door switch		

1 The main switch is the odd one out. Explain why it has to be connected differently from the three appliances.

2 The heater draws a lot of current. Explain why adding it to the room makes a fuse more important, not less.

CHALLENGE

Some modern holiday light strings are wired in series but keep working when one bulb burns out. Each bulb contains a small extra part called a shunt that starts conducting when the filament fails. Explain, using the idea of the loop, exactly what that shunt must be doing.

Answer Key: Closing the Loop

QUESTION PAGE

1. The loop must be complete. There must be an unbroken conducting path from one terminal of the battery, through the components, and back to the other terminal.

WHAT YOU DO	RESULT	WHY
Remove L1 from A	Both lamps go out	Series has one path. Removing a lamp breaks the only loop.
Remove L3 from B	L4 stays lit, same brightness	Parallel branches are independent. The other branch is still a complete loop.
Open a switch in A	Everything goes out	A gap anywhere in a single-path loop stops all current.
Add a third lamp in series to A	All three dim further	The battery's push is now shared three ways instead of two.

3. **B, the parallel circuit.** In series the battery's push is shared between the lamps so each gets a fraction. In parallel each branch receives the full push, so each lamp is as bright as it would be on its own.

4. Old light strings are wired in series, so one burnt-out bulb breaks the single loop and everything goes dark. A house is wired in parallel, so each light is on its own branch and one has no effect on another.

5. The fuse is designed to break first, at a safe and predictable point, before the current gets high enough to overheat the wiring inside the walls. Replacing a fuse is trivial; replacing burnt wiring or recovering from a fire is not.

GO FURTHER PAGE

Table: All three appliances in **parallel**, so each gets the full supply and each can be switched independently. The main door switch in **series** with the whole room, so opening it breaks the single path that feeds everything.

1. The appliances must work independently, which needs parallel. The main switch must kill everything at once, which needs it to sit in the one path that all the branches depend on.

This is the best question on the page. It requires holding both arrangements in mind at once and choosing based on purpose.

2. A heater draws a large current. Parallel branches add together, so total current through the shared supply wiring rises and the risk of overheating goes up. The fuse is the thing that breaks the loop before that happens.

Challenge: When the filament fails it creates a gap in the series loop. The shunt must close that gap by providing an alternative conducting path through the dead bulb, so the loop is complete again and the rest of the string stays lit.

Full credit for any answer built on "it re-completes the loop". Strong students note the remaining bulbs get slightly brighter, since the push is now shared among fewer working bulbs.

Same Moon, Different Shape

TEACHER BRIEFING • KEEP THIS PAGE

NO SCIENCE BACKGROUND NEEDED. NO LAB, NO EQUIPMENT.

Everything you need is on this page and the answer key. Nothing in this lesson is heated, poured, cut or plugged in.

Copy for students: the next three pages. **Keep:** this page and the answer key.

What this lesson does

Two of the most confidently held wrong ideas in science live in this lesson: that moon phases are Earth's shadow, and that seasons are caused by distance from the Sun. Both are addressed with evidence a student can check rather than with assertion.

How the period runs

MIN	WHAT YOU DO
0-5	Read the opening lines below. Hand out all three student pages.
5-14	Students read the page.
14-40	Question page.
40-53	Go Further: the two hemispheres problem, which settles the seasons question on its own.
53-55	Collect all pages.

SAY THIS TO START

Two questions, and I will warn you now that most adults get both wrong. Why does the Moon change shape? And why is it hotter in summer? The popular answers are Earth's shadow and being closer to the Sun. Both are wrong, and the evidence that kills them is something you could check yourself this week.

The three questions students will ask

THEY ASK	YOU SAY
"Isn't a moon phase Earth's shadow on the Moon?"	No. That is a lunar eclipse, which is rare and lasts a couple of hours. Phases happen every month and depend on which lit part of the Moon we can see from our angle.
"Isn't summer when we're closer to the Sun?"	No. Earth is actually slightly closer to the Sun in January. It is summer somewhere on Earth at all times of year, which distance could never explain.
"Does the Moon have a dark side?"	There is a far side we never see from Earth, but it gets just as much sunlight as the near side. It is far, not dark.

FINISHED EARLY?

The Go Further page uses the fact that Australia has summer in December to disprove the distance explanation in a single step.

Read: Two Things Everyone Gets Wrong

Two Things Everyone Gets Wrong

Written for this lesson

- 1 Half the Moon is always lit by the Sun, just as half of Earth is always in daylight. That never changes. What changes is **how much of the lit half we can see from where we are standing**.
- 2 The Moon takes about 29.5 days to orbit Earth. When it sits roughly between us and the Sun, the lit half faces away from us and we see almost nothing: a new moon. When Earth sits between the Sun and the Moon, the lit half faces straight at us and we see a full moon. In between we see a slice, and the slice grows and shrinks as the angle changes.
- 3 The popular explanation is that phases are Earth's shadow falling on the Moon. Here is why that cannot be right. Earth's shadow does sometimes fall on the Moon, and when it does we call it a lunar eclipse. Lunar eclipses happen a couple of times a year at most and are over in a few hours. Phases happen every single month, on a strict schedule, and take two weeks to go from new to full. They are not the same thing and they do not happen on the same timetable.
- 4 There is a second clue. During a crescent moon you can often see the rest of the Moon faintly, lit by sunlight bouncing off Earth. If Earth's shadow were falling across it, that part would be genuinely dark, not dimly lit.
- 5 Now the seasons. The popular explanation is that summer happens when Earth is closer to the Sun. Earth's orbit really is slightly oval, so the distance does change a little through the year. But the explanation collapses on one fact: Earth is at its **closest** to the Sun in early January, which is the middle of winter in the northern hemisphere.
- 6 It collapses harder on a second fact. When it is winter in New York it is summer in Sydney, on the same day, with Earth at exactly one distance from the Sun. Distance is a single number for the whole planet at any moment. It cannot possibly produce opposite seasons in two places at once.
- 7 The real cause is **tilt**. Earth's axis is tipped about 23.5 degrees and stays pointing the same way as we orbit. For half the year the northern hemisphere leans toward the Sun; for the other half it leans away. The southern hemisphere does the opposite at the same time.
- 8 Leaning toward the Sun does two things at once. Sunlight arrives at a steeper angle, so the same amount of energy is concentrated onto a smaller patch of ground instead of being smeared across a wide one. And the days are longer, so that concentrated energy arrives for more hours. More energy per patch, for more hours per day, is what summer is.

Questions: Same Moon, Different Shape

1 How much of the Moon is lit by the Sun at any moment? What changes to give us different phases?

2 Give two separate reasons from the text why moon phases cannot be Earth's shadow.

3 Earth is closest to the Sun in early January. Why is that a problem for the idea that summer is caused by being closer?

4 Explain the two things that happen when a hemisphere leans toward the Sun. Both matter.

5 The Moon has a far side we never see from Earth. Is it correct to call it the dark side? Explain.

6 New moon to full moon takes about 15 days. A lunar eclipse lasts a few hours. What does that difference tell you?

Go Further: Same Day, Opposite Seasons

THE DATA

All figures recorded on the **same day**, 21 December.

CITY	HEMISPHERE	DAYLIGHT HOURS	AVG TEMP	SEASON
Chicago, USA	Northern	9.1	28 F	Winter
Quito, Ecuador	On the equator	12.1	66 F	Neither
Sydney, Australia	Southern	14.4	77 F	Summer

- 1

On 21 December, how far is Chicago from the Sun compared with Sydney? Think carefully before answering.
- 2

Use your answer to question 1 to explain why distance from the Sun cannot be the cause of the seasons.
- 3

Quito is on the equator and gets about 12 hours of daylight every day of the year. Using the idea of tilt, explain why the equator does not really have seasons.

CHALLENGE

Earth's tilt is 23.5 degrees. Imagine a planet with no tilt at all, orbiting its star in a perfect circle. Describe what the seasons would be like there, and what would happen to daylight hours through the year.

Answer Key: Same Moon, Different Shape

QUESTION PAGE

1. Always exactly half, and that never changes. What changes is how much of the lit half is visible from our position as the Moon orbits.
2. Any two of: lunar eclipses happen only a couple of times a year while phases happen every month; an eclipse lasts hours while a phase cycle takes weeks; during a crescent you can often see the rest of the Moon faintly lit by light reflected from Earth, which would be genuinely dark if it were in shadow.
3. Because January is the middle of northern winter. If closer meant hotter, January should be the hottest month in the north, and it is the coldest.
4. 1. Sunlight strikes at a steeper angle, so the same energy is concentrated onto a smaller area of ground. 2. Days are longer, so that concentrated energy arrives for more hours.
Both halves are needed for full credit. Most students give the daylight-hours half only.
5. No. The far side receives just as much sunlight as the near side. It is far from us, not dark.
6. They are completely different events on completely different timescales, so one cannot be the explanation for the other.

GO FURTHER PAGE

1. **Essentially the same distance.** Both cities are on the same planet, and the diameter of Earth is trivial compared with the 93 million miles to the Sun.
This is the question that does the work. Some students answer "Sydney is closer." Ask them how big Earth is compared with the distance to the Sun.
2. Because the two cities are the same distance from the Sun on the same day and are having opposite seasons. Distance is one number for the whole planet at any moment, so it cannot produce two different results at once.
3. The equator is never tilted strongly toward or away from the Sun, so the angle of sunlight and the length of the day stay nearly constant all year. Without that variation there is nothing to produce seasons.

Challenge: No seasons at all. Every latitude would receive the same angle of sunlight and the same number of daylight hours every day of the year. Everywhere would have twelve hours of daylight permanently, and each place would have one steady climate.
Strong students note the poles would stay permanently cold and the equator permanently hot, just without any change through the year.

Reading the Rocks

TEACHER BRIEFING • KEEP THIS PAGE

NO SCIENCE BACKGROUND NEEDED. NO LAB, NO EQUIPMENT.

Everything you need is on this page and the answer key. Nothing in this lesson is heated, poured, cut or plugged in.

Copy for students: the next three pages. **Keep:** this page and the answer key.

What this lesson does

Students order five rock layers oldest to youngest, then use the fossils in them to reconstruct what the environment used to be. The lesson teaches two rules that geologists actually use, then immediately makes students apply them to a cross-section that includes an intrusion, which is where the second rule earns its keep.

How the period runs

MIN	WHAT YOU DO
0-6	Read the opening lines below. Hand out all three student pages. Point at the diagram while reading the two rules.
6-14	Students read the page and study the cross-section.
14-40	Question page.
40-53	Go Further: what the fossils reveal about the past environment.
53-55	Collect all pages.

SAY THIS TO START

This is a lesson about reading. Not reading words, reading rock. A cliff face is a stack of pages, and there are two rules that let you put them in order. Rule one: in undisturbed layers, the bottom is oldest, because it had to be there first for anything to land on it. Rule two: anything that cuts through a layer must be younger than the layer it cuts, because you cannot cut something that does not exist yet.

The three questions students will ask

THEY ASK	YOU SAY
"How do they know how old it actually is?"	This lesson is about relative age, which is putting events in order. Actual ages in years come from radioactive dating, which is a different method.
"How can there be sea fossils on a mountain?"	Because that rock formed underwater and was later pushed up. Rock moves over millions of years. It is genuinely one of the most surprising things in science.
"What is an intrusion?"	Molten rock that pushed up through existing layers and then cooled solid. It cuts across the layers instead of sitting flat on top of them.

FINISHED EARLY?

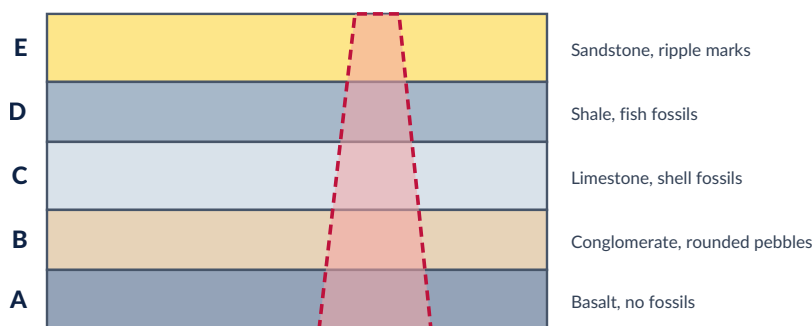
The Go Further page has students reconstruct the environment from the fossil evidence in each layer, which is a genuinely satisfying puzzle.

Read: The Cliff Face

THE TWO RULES

Rule 1. In undisturbed layers, the **bottom is oldest**. Each layer settled on top of the one below, so the stack records the order it happened in.

Rule 2. Anything that **cuts through** a layer must be **younger** than the layer it cuts. You cannot cut through something that does not exist yet.



Dashed body = granite intrusion. It is NOT one of the lettered layers.

Layers are shown as they were found, undisturbed.

- 1 The diagram shows a cliff face. Layers **A** to **E** are sedimentary rock, formed when sand, mud or shell fragments settled, piled up and were compressed over long periods. Each layer took a very long time to form and settled on top of whatever was already there.
- 2 The dashed shape is different. It is an **intrusion**: molten rock that forced its way up through the existing layers and cooled solid in place. It does not lie flat like the others. It cuts across them.
- 3 That difference is what rule 2 is for. The intrusion sits low in the picture, so rule 1 would wrongly suggest it is old. But it cuts upward through layers above it, and it can only have cut through rock that was already there. So it is younger than every layer it passes through, even though it appears near the bottom.
- 4 The fossils in each layer tell a second story. Fossils form when a living thing is buried quickly enough to be preserved, which happens far more often underwater than on land. Shell fossils in a limestone layer mean that layer formed on a sea floor. Ripple marks in a sandstone layer mean shallow moving water, like a beach or a riverbed.
- 5 Put those together and a cliff becomes a record of how one place changed. If a layer with deep-sea shells sits below a layer with beach ripple marks, that place was once deep ocean and later became shoreline. The rock did not travel. The environment changed around it, or the land itself rose.
- 6 None of this gives an age in years. It gives an **order**, which is called relative dating. Actual ages come from measuring radioactive elements inside the rock, which is a separate technique. Order alone is still remarkably powerful: it is how the entire sequence of life on Earth was worked out before anyone could date a rock at all.

Questions: Reading the Rocks

1 Using rule 1, list layers A to E in order from oldest to youngest.

2 Where does the intrusion belong in that order? Explain which rule you used and why rule 1 would have misled you.

3 Layer C contains shell fossils. What does that tell you about what this place was like when layer C formed?

4 Layer E contains ripple marks. What kind of environment makes those, and what does that tell you about how the place changed between C and E?

5 Layer A contains no fossils at all. Suggest a reason why a rock layer might contain no fossils.

6 Explain the difference between relative dating and knowing a rock's age in years.

TASK

Using the fossil and texture evidence in each layer, reconstruct what this place was like at each stage. Work from the bottom up, oldest first.

LAYER	EVIDENCE FOUND IN IT	WHAT THIS PLACE WAS LIKE THEN
A	Basalt, no fossils	
C	Limestone, shell fossils	
D	Shale, fish fossils	
E	Sandstone, ripple marks	

1 Write the story of this place in three or four sentences, in order, from oldest to youngest.

2 This cliff is now 1,200 feet above sea level, yet layers C and D formed underwater. Give one explanation for how that could happen.

CHALLENGE

A geologist finds the **same** three fossil species, in the same order, in cliffs on two different continents. Explain why that is useful, and what it lets her conclude about the two sets of rock.

Answer Key: Reading the Rocks

QUESTION PAGE

1. Oldest to youngest: **A, B, C, D, E**. A is at the bottom, so it was there first.
2. The intrusion is **younger than every layer it cuts through**, so it belongs at or near the top of the order despite appearing low in the diagram. Rule 2 gives this. Rule 1 would have misled you because it only works for layers that settled flat and undisturbed, and an intrusion did not settle at all.
This is the question worth marking closely. Students who apply rule 1 to the intrusion have not understood why there are two rules.
3. It formed on a sea floor. Shell fossils and limestone both point to marine conditions.
4. Ripple marks form in shallow moving water such as a beach or riverbed. So between C and E the place changed from open sea to shallow water or shoreline, meaning the water became shallower or the land rose.
5. Any reasonable answer: basalt is cooled lava, and nothing living survives being buried in molten rock; or no organisms lived there; or conditions did not allow preservation.
6. Relative dating puts events in order without saying how long ago. Absolute age in years comes from measuring radioactive elements. Order tells you what happened first; it does not tell you when.

GO FURTHER PAGE

LAYER	ENVIRONMENT THEN
A	Volcanic. Lava flowed and cooled here.
C	A sea floor. Shelled marine animals lived and died here.
D	Still underwater, quiet enough for fine mud to settle and preserve fish.
E	Shallow moving water, a beach or riverbed.

1. Something like: this place began as a volcanic landscape. It was later covered by sea, where shelled animals and then fish lived and were buried in mud. The water became shallower until it was a beach or riverbed. Later still, molten rock forced its way up through all of it.

Reward any account that runs in the right order and uses the evidence. Do not require the intrusion to be mentioned, but note it if a student includes it.

2. The land was pushed upward over a very long time, by plate movement or mountain building. Accept sea levels falling as an alternative.

Challenge: The same fossils in the same order means the two sets of rock formed at roughly the same times as each other, so she can match them up across continents even without knowing their ages in years.

This is index fossils, a real and important technique. Strong students may add that it suggests the two regions shared conditions or were once closer together.

Emergency Day Checklist

PRINT THIS PAGE AND LEAVE IT ON TOP

Fill in the blanks once, at the start of the year, and this page works for every absence after that.

NOTE FOR THE SUBSTITUTE

My name _____

Room / period _____

Teach this lesson Lesson # ____ (pages ____ to ____) or any lesson from the index

Teacher next door _____

Lab / storeroom policy Keep it closed. Nothing in this pack needs equipment.

Students who leave early _____

Students with accommodations _____

Seating, extra time, reader _____

Fire drill meeting point _____

Leave the work _____

Anything else _____

SUBSTITUTE: BEFORE YOU LEAVE, DO THESE THREE THINGS

1. Stack the collected work on the teacher's desk with this page on top.
2. Write the names of two or three students who were especially helpful. Teachers genuinely act on this.
3. Note anything the teacher should know, in one line. Not a report. One line.

SUBSTITUTE'S NOTE BACK TO THE TEACHER

Lesson taught _____

Got through All of it / Most of it / Page 1 only

Helpful students _____

Anything to know _____

REUSE NOTE

This pack is designed to be printed once and reused across many absences. Store the briefing pages and answer keys separately from the student pages so the wrong sheets never get photocopied.

NGSS MIDDLE SCHOOL • GRADES 6-8

Each lesson is mapped to the closest matching middle school performance expectation. Most lessons also touch others; the primary alignment is listed for clarity.

#	LESSON	FOCUS	NGSS
11	Something Changed	Physical vs chemical change, conservation of mass	MS-PS1-2
12	Good Vibrations	Waves: amplitude, wavelength, frequency	MS-PS4-1
13	Where Traits Come From	Heredity, alleles, Punnett squares	MS-LS3-2
14	The Beetles That Survived	Natural selection and adaptation	MS-LS4-4
15	Trading Gases	Photosynthesis and cellular respiration	MS-LS1-6
16	The Delivery Network	Circulatory and respiratory systems	MS-LS1-3
17	Why It Floats	Density and buoyancy	MS-PS1-1
18	Closing the Loop	Circuits, series and parallel	MS-PS3-5
19	Same Moon, Different Shape	Moon phases and the seasons	MS-ESS1-1
20	Reading the Rocks	Rock cycle, strata, relative dating	MS-ESS1-4

Coverage by domain

DOMAIN	LESSONS	WHAT IS COVERED
Physical Science	11, 12, 17, 18	Physical and chemical change, conservation of mass, wave properties, density and buoyancy, series and parallel circuits
Life Science	13, 14, 15, 16	Inheritance and Punnett squares, natural selection, photosynthesis and respiration, body systems
Earth and Space Science	19, 20	Moon phases, the cause of seasons, rock layers and relative dating

HOW THIS PACK RELATES TO VOLUME 1

Volume 1 covers fair tests and variables, cells, forces and motion, the periodic table, energy transfer, food webs, the particle model, the water cycle and weather, plate tectonics, and evaluating data and claims.

No topic is repeated between the two volumes. Together they give twenty emergency science lessons with no overlap.

This resource is independently created and is not affiliated with, endorsed by, or reviewed by any state agency or standards body. NGSS codes are provided to help teachers document the day's instruction.

Terms of Use and Credits

THANK YOU

Thank you for buying Volume 2. It was written by a practicing teacher with more than twenty years in secondary classrooms, and every lesson is built the way I would want a sub plan built if I were the one waking up sick.

If it saved you a morning, a rating on the listing genuinely helps other teachers find it, and earns you credit toward your next purchase.

Getting the most out of this pack

- **Print it before you need it.** The whole point is that it works on a morning when you cannot get out of bed to print anything.
- **Store it in two piles.** Student pages in one, briefing pages and answer keys in the other, so the wrong sheets never get copied by mistake.
- **Tell your office where it is.** On a genuine emergency day, the person who needs to find this is not you.
- **Use them when you are present too.** These work as standalone skill lessons, review days, and non-lab days when the lab is booked out.

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OTHER PACKS IN THIS SERIES

Emergency Sub Plans · Middle School Science, Volume 1 · ten different lessons, no lab required, 57 pages

Emergency Sub Plans · Middle School Math · ten no-prep lessons, 57 pages

Emergency Sub Plans · Middle School ELA · ten no-prep lessons, no novel required, 57 pages

Emergency Sub Plans · Computer Science and Digital Technologies · 57 pages

Available together as a discounted bundle. Fill the drawer once.

Credits

Written, designed and typeset by Clarence Goh. Set in Lato and JetBrains Mono.

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